

ATG n8n Sidecar — Pre-class Setup

Before you start: the service-setup-checklist is complete. You already have:

- A development environment (Python 3.11+, Node 18+, VS Code)
- An Amzur LiteLLM virtual key (LITELLM_API_KEY in your chatbot .env)
- A Supabase project with your chatbot database running (DATABASE_URL in your .env)
- A Google Cloud project with OAuth credentials for chatbot login (GOOGLE_CLIENT_ID, GOOGLE_CLIENT_SECRET)
- A Google Service Account for Sheets (GOOGLE_SERVICE_ACCOUNT_JSON)

This guide adds **only what is needed for the n8n sidecar session**. Do not recreate anything already set up.

Time required: 20–30 minutes.

If you get stuck: post in the training Slack channel.

What You Need Ready

By the end of this setup you should have:

- An n8n Cloud instance at <https://<yourname>.app.n8n.cloud>
- LiteLLM (training) credential in n8n (uses your existing virtual key)
- Postgres (training) credential in n8n connected to your existing Supabase project
- Gmail (training) credential in n8n (uses your existing Google Cloud project)
- A tickets table created in your Supabase database
- A verify_setup workflow confirming n8n can call the LLM

Part 1 — Create n8n Cloud Account

1. Go to **n8n.io** → **Get started for free**
2. Sign up with an email you can access during class
3. After signup, your instance URL will be: <https://<yourname>.app.n8n.cloud>
4. Complete the owner account setup when prompted
5. Confirm you land on the **Overview** page (shows Workflows and recent activity)

Note: Use a personal Gmail if possible. Corporate Google Workspace accounts sometimes block OAuth for external apps.

Part 2 — Add LiteLLM Credential in n8n

You already have an Amzur LiteLLM virtual key from the service-setup-checklist. You just need to register it in n8n.

Use the values from your existing chatbot .env:

n8n Field	Value from your .env
API Key	LITELLM_API_KEY
Base URL	LITELLM_PROXY_URL (must end with /v1)

Steps:

1. Click the **+** button at the top of the left sidebar (next to the n8n logo)
2. Hover **Credential** → click **Personal**
3. Search **OpenAI** → select **OpenAI**
4. Fill in the fields:
 - **API Key:** paste your LITELLM_API_KEY
 - **Base URL:** paste your LITELLM_PROXY_URL — confirm it ends with /v1 (e.g. `https://litellm.amzur.com/v1`)
5. Leave **Organization ID** blank
6. Click **Save** → name it LiteLLM (training)
7. Confirm the credential test turns **green**

Note: If the Base URL field is blank or missing, n8n will call OpenAI directly and fail. The Base URL is required.

Part 3 — Add Supabase PostgreSQL Credential in n8n

Your chatbot app connects directly to Supabase on port 5432. n8n Cloud needs to use the **Transaction Mode connection pooler on port 6543** — this handles the many short-lived connections from a serverless environment without exhausting Supabase's connection limit.

3.1 — Get the pooler connection string from Supabase

1. Open your Supabase project dashboard
2. Click the **Connect** button in the top bar (next to the branch name)
3. In the Connect dialog, select the **Connection string** tab → choose **Transaction** mode
4. Copy the connection string:
`postgresql://postgres.<project-ref>:<password>@aws-0-<region>.pooler.supabase.com:6543/postgres`

Note: The Database → Settings page only shows pool size configuration — the actual connection strings are only in the Connect dialog.

3.2 — Create the credential in n8n

1. Click **+** → hover **Credential** → **Personal** → search **PostgreSQL** → select **PostgreSQL**
2. Fill in the fields using the pooler connection string:

Field	Value
Host	aws-0-<region>.pooler.supabase.com
Port	6543
Database	postgres
User	postgres.<project-ref>

Field	Value
Password	your Supabase database password
SSL	Disable

3. Click **Save** → name it Postgres (training)
4. Click **Test connection** — confirm it passes

⚠ **Port 6543 only.** Do not use port 5432 (direct connection) for n8n. The pooler is required for n8n Cloud's connection pattern.

Part 4 — Enable Gmail in Your Google Cloud Project

You already have a Google Cloud project from the chatbot setup. You need to enable the Gmail API and add an n8n redirect URI to the existing OAuth client.

4.1 — Enable Gmail API

1. Go to **console.cloud.google.com**
2. Select your existing **amzur-chatbot-dev** project
3. Go to **APIs & Services** → **Library**
4. Search **Gmail API** → click it → **Enable**

4.2 — Add n8n redirect URI to your existing OAuth client

1. Go to **APIs & Services** → **Credentials**
2. Click the Web Application OAuth client you created for the chatbot
3. Under **Authorised redirect URIs**, click **+ Add URI** and add:
`https://<yourname>.app.n8n.cloud/rest/oauth2-credential/callback`
4. Click **Save**

Note: Your email must already be in the Test Users list from the earlier Google OAuth setup. If not, add it under **OAuth consent screen** → **Test users**.

4.3 — Add Gmail credential in n8n

1. Click **+** → hover **Credential** → **Personal** → search **Gmail** → select **Gmail OAuth2**
2. Paste the same **Client ID** and **Client Secret** from your Google Cloud OAuth client
3. Click **Sign in with Google** → authorize with your Gmail account
4. Click **Save** → name it Gmail (training)

Note: If Gmail OAuth fails (e.g. `redirect_uri_mismatch`), double-check the redirect URI in step 4.2 exactly matches your n8n instance URL. If it still fails, continue to Part 5 — Gmail can be set up during the session without blocking the rest.

Part 5 — Create the tickets Table in Supabase

The n8n sidcar stores tickets in your Supabase database. Create the table now so it exists when you test.

1. In your Supabase project, click **SQL Editor** in the left sidebar
2. Click **+ New query**
3. Paste and run:

```
CREATE TABLE IF NOT EXISTS tickets (  
  id          uuid      PRIMARY KEY DEFAULT gen_random_uuid(),  
  ticket_id   text      UNIQUE NOT NULL,  
  user_email  text      NOT NULL,  
  issue       text      NOT NULL,  
  category    text      NOT NULL DEFAULT 'General',  
  priority    text      NOT NULL DEFAULT 'medium',  
  status      text      NOT NULL DEFAULT 'open',  
  created_at  timestamptz NOT NULL DEFAULT now(),  
  updated_at  timestamptz NOT NULL DEFAULT now(),  
  thread_id   text,  
  assigned_team text,  
  next_action text  
);
```

4. Click **Table Editor** in the left sidebar — confirm tickets appears in the table list

⚠ **Important:** This table is **not** created by the existing Alembic migrations. You must create it manually here. Without it, the first ticket creation call will return a 502 error.

Part 6 — Add n8n Environment Variables to Your App

The actual webhook URLs are generated during the session, but add the variable names to your .env now so the app boots cleanly before the session.

Open your chatbot .env and add at the bottom:

— n8n automation sidcar (fill in after activating workflows in session) —

N8N_WEBHOOK_URL=

N8N_API_KEY=

N8N_STATUS_WEBHOOK_URL=

Leave the values blank. The app boots without them. Do the same in .env.example so the variable names are documented.

Part 7 — Verify n8n + LiteLLM

Build a small workflow to confirm n8n can reach the model before the session.

1. Click the **+** button at the top of the left sidebar → hover **Workflow** → click **Personal**
2. Click the workflow title at the top → rename it **verify_setup**
3. Click **+** on the canvas → search **Chat Trigger** → select it
4. Click **+** on the Chat Trigger output → search **AI Agent** → select it
5. In the AI Agent node: set **Source for Prompt** to **Connected Chat Trigger Node**
6. Click **+** on the AI Agent's **Chat Model** port → search **OpenAI Chat Model** → select it
 - **Credential:** LiteLLM (training)

- **Model:** gpt-4o
 - Expand **Options** → set **Use Responses API** to **OFF**
7. Click **Save** (Ctrl+S)
 8. Click the **Chat** button at the bottom of the canvas → send: *Hello, are you working?*
 9. A meaningful reply confirms n8n + LiteLLM is ready.
- Note:** If the model hangs or errors, confirm gpt-4o is available on the training LiteLLM key. Ask your trainer.

Part 8 — Optional: Add n8n Docs MCP Server in VS Code

This step is optional but recommended. It gives you direct access to n8n documentation inside VS Code through an MCP server, so you can quickly ask documentation-based questions while building workflows.

This does **not** replace n8n Cloud, LiteLLM, Supabase, or Gmail setup. It is only a documentation helper for learners while they are working on the chatbot and n8n sidecar integration.

Why we are adding this

During the n8n sidecar session, learners may need to quickly check:

- How a specific n8n node works
- What configuration a node expects
- How webhooks, credentials, HTTP Request, Gmail, Postgres, or AI Agent nodes are used
- How to troubleshoot common n8n workflow issues
- How to understand n8n concepts without switching repeatedly between browser tabs

The n8n Docs MCP server helps bring n8n documentation closer to the development environment.

Add MCP configuration in VS Code

In your project, create or update the following file:

.vscode/mcp.json

Add this configuration:

JSON

```
{
  "servers": {
    "n8n-docs": {
      "type": "http",
      "url": "https://n8n.mcp.kapa.ai/"
    }
  },
  "inputs": []
}
```

Save the file.

Verify the MCP server is running

After saving the file, VS Code should show the MCP server as:

Running | Stop | Restart | tools | More...

Pre-session Checklist

Required (must be done before the session)

- n8n Cloud opens at your instance URL (<https://<yourname>.app.n8n.cloud>)
- LiteLLM (training) credential is saved — credential test is green
- Postgres (training) credential connects to Supabase on port 6543 — test connection passes
- tickets table visible in Supabase **Table Editor**
- n8n env vars added to .env (values blank is fine — app must boot without errors)
- verify_setup workflow returns a model response

Recommended (can be completed during session if needed)

- Gmail (training) credential is authorized in n8n
- Gmail API is enabled in your Google Cloud project
- n8n OAuth redirect URI is added to your Google Cloud OAuth client

What to Add to Your .env

All your existing variables stay unchanged. Add only these three lines at the bottom of your .env and .env.example:

n8n automation sidecar (fill in after activating workflows in session)

N8N_WEBHOOK_URL=

N8N_API_KEY=

N8N_STATUS_WEBHOOK_URL=

Leave the values blank for now. The app boots without them.

Troubleshooting

Symptom	Fix
LiteLLM credential test fails	Check API key and Base URL. The Base URL must end with /v1 (e.g. https://litellm.amzur.com/v1). Confirm you are using the training key from your .env.
Can't find the credential menu	Click the + button at the top-left → hover Credential → Personal . To view existing credentials, go to Overview → Credentials tab.
OpenAI credential type not found	Search just "OpenAI" — the credential is listed as OpenAI (not "OpenAi account").
Chat model hangs in verify_setup	Confirm gpt-4o is enabled on the training LiteLLM key. Turn Use Responses API OFF on the OpenAI Chat Model node.
PostgreSQL connection refused	Confirm you are using port 6543 (pooler) not 5432 (direct). Set SSL to Disable .
self-signed certificate in certificate chain	Set SSL to Disable in the PostgreSQL credential. Disabling SSL is safe here — Supabase does not enforce SSL on incoming pooler connections.
PostgreSQL auth fails	Re-copy the password from the Supabase pooler connection string — it may be URL-encoded.
tickets table not found in Table Editor	Run the Part 5 SQL in the SQL Editor. The table is not created by Alembic.
Gmail redirect_uri_mismatch	The n8n redirect URI must be added to the same OAuth client used by the chatbot. Copy the URI exactly: <a href="https://<yourname>.app.n8n.cloud/rest/oauth2-credential/callback">https://<yourname>.app.n8n.cloud/rest/oauth2-credential/callback

Symptom	Fix
Gmail app blocked	Your email must be in the Test Users list under OAuth consent screen → Test users . Add it if it's missing.
Failed to parse. Expected object, received array	Turn Use Responses API OFF on the OpenAI Chat Model node.
AI Agent answers but ignores chat input	Set AI Agent Source for Prompt to Connected Chat Trigger Node .

If the required checklist is complete, you are ready for the session. The Gmail credential can be set up during the session if needed.